

EXHIBIT E-3.35

GALLIUM TRUE-SOURCE DEPENDENCY ASSESSMENT (TSDA)

Analysis Dataset and Preliminary Assessment

Researcher: Blessing Ameh | Product: Gallium metal (HTS 8112.92.1000) | Comparison: 2019-2022 vs. 2023-2025

1. Purpose and Principal Finding

This exhibit applies TSDA to U.S. gallium-metal sourcing before and after China's 2023 gallium export controls. The principal question is whether changes in the countries directly supplying the United States corresponded to a robust reduction in underlying foreign dependence across direct supply, production, processing, and intermediary routes.

Principal finding. U.S. direct-supplier concentration did not show a robust decline. The pooled direct-supplier HHI increased from 1,557.5 to 2,416.9, while the dominant supplier share rose from 25.65% to 40.01%. Under TSDA Methodology Version 1.1, the gallium direct-supplier result is No Diversification. Upstream production concentration also increased.

Direct U.S. supplier results

Measure	2019-2022	2023-2025	Change
Total U.S. import quantity	30,412 kg	46,503 kg	+52.91%
Direct-supplier HHI	1,557.5	2,416.9	+859.4
Dominant supplier	Japan - 25.65%	Canada - 40.01%	+14.36 pp
Direct China share	18.04%	8.67%	-9.37 pp
Annualized direct-China quantity	1,371 kg/yr	1,344 kg/yr	-2.0%
Annual-HHI sensitivity	Avg. 2,807.5	Avg. 2,641.0	Volatile/mixed; no sustained decline

The decline in China's import share should not be read as an equivalent physical contraction of the direct China route: annualized quantity was nearly unchanged while total U.S. imports expanded substantially.

Supplier	2019-2022 share	2023-2025 share
Canada	8.53%	40.01%
Germany	18.83%	21.19%
Japan	25.65%	15.02%
China	18.04%	8.67%
Taiwan	8.88%	6.10%
Other countries	20.07%	9.02%

2. Upstream Production and Intermediary Tracing

TSDA then looks beyond the country recorded at the U.S. border. The gallium application shows that several major alternative suppliers cannot be treated automatically as independent upstream sources because production, refining, recycling, or feedstock provenance remains unresolved.

Primary-production concentration

Measure	2019-2022	2023-2025	Result
Production HHI - principal revised series	9,498.0	9,751.3	Higher
China share of named primary production	97.44%	98.74%	+1.30 pp
Production HHI - first-published sensitivity	9,449.4	9,730.8	Higher
U.S. net import reliance	100%	100%	Unchanged

Intermediary-tracing findings

Route	Evidence	TSDA treatment
Canada	Secondary/refining source; mixed customs code; scrap/feedstock provenance unresolved.	Do not attribute proportionally to China.
Germany	Gallium-specific Eurostat data show substantial China exposure, but Germany is a refining/trading node.	U.S.-bound upstream provenance unresolved; transformation rule applies.
Japan	Domestic production, refining, recycling, and material China-linked imported feedstock coexist.	Mixed origin; exact U.S.-bound China share unresolved.
Taiwan	Most detailed CCC line is a mixed residual code rather than gallium-specific.	Traced procedurally; gallium provenance unresolved.
Russia	USGS identifies Russia as a primary producer; domestic output materially exceeds the U.S.-bound route.	Preserved as Russia in the conservative China-max scenario; shipment-level provenance not claimed.
Slovakia	High-purity refining and secondary recovery source; upstream low-purity/recycled feedstock unresolved.	Included in 5% sensitivity screen.

Conservative attribution scenarios

Period	Scenario A HHI	Scenario B HHI	Interpretation
2019-2022	1,557.5	8,886.3	Unresolved branches separate vs. unresolved non-Russia branches attributed to China.
2023-2025	2,416.9	9,819.3	Both scenario endpoints increase.

Important limitation: these are attribution scenarios, not formal mathematical identification bounds and not a point estimate of true-source HHI. A direct supplier may itself contain multiple upstream sources.

3. Empirical Validation Against the Post-Shock Response

The original Section 8 validation rule is retained unchanged for this initial application. Routes that were not defensibly classified before the shock remain exploratory rather than being reclassified after the outcome is observed.

Route	Annualized quantity change	Unit-value change	Validation treatment
China	-2.0%	+14.6%	Reference route; not an intermediary prediction.
Germany	+129.4%	+149.9%	Exploratory only; pre-shock upstream provenance unresolved.
Canada	+856.3%	+81.3%	Exploratory only; scrap/feedstock provenance unresolved.
Japan	+19.4%	+73.0%	Exploratory only; mixed upstream sources.
Taiwan	+40.0%	+254.8%	Exploratory only; mixed customs code.
Russia	-68.6%	+30.7%	Raw quantity prediction contradicted, but materially confounded by a Russia-specific U.S. tariff shock.
Slovakia	+218.5%	+15.6%	5% sensitivity route; exploratory.
All U.S. imports	+103.9%	+72.0%	Market benchmark.

Russia confounder confirmed. Public Law 117-110 made HTS column 2 rates applicable to Russian goods effective April 9, 2022. The 2022 HTS entry for gallium (8112.92.10.00 / statistical code 8112.92.1000) lists a 3% column 1-general rate and a 25% column 2 rate. The 22-percentage-point tariff increase is a material non-gallium explanation for the post-2022 contraction of the Russia route.

Validation finding. The Section 8 result is mixed and should not be overstated. Russia is the only route that met the pre-shock condition for a formal independent-source test, and its raw quantity outcome is contradicted but materially confounded by the 2022 tariff change. Germany, Canada, Japan, Taiwan, and Slovakia provide useful descriptive post-shock evidence but were not sufficiently resolved before the shock for a clean binary validation test.

Market-relative unit values are retained as an exploratory sensitivity only. The original pre-specified validation rule is not rewritten after observing the gallium outcome.

4. Preliminary TSDA Assessment

TSDA component	Observed result	Assessment
Direct-supplier concentration	HHI 1,557.5 -> 2,416.9; dominant share 25.65% -> 40.01%.	No robust decline.
Primary-production concentration	HHI 9,498.0 -> 9,751.3; China share 97.44% -> 98.74%.	Upstream concentration increased.
Processing / refining	Important processing nodes identified, but consistent country-level quantitative shares are incomplete.	Quantitative processing concentration unresolved.
True-source attribution	Major routes traced, but several U.S.-bound feedstock origins remain unresolved.	Report scenarios, not a point estimate.
Net import reliance	100% in both periods.	Foreign reliance remained complete.
Empirical validation	Formal Russia test confounded; other routes exploratory.	Mixed / not conclusive.

Preliminary conclusion. The gallium case does not show a robust reduction in U.S. direct-supplier concentration after the shock. The largest direct supplier changed from Japan to Canada, and China's recorded share fell, but primary production became even more concentrated in China and major alternative supplier routes remain partly unresolved upstream. Accordingly, the available evidence does not establish that the United States achieved a deeper structural reduction in foreign gallium dependence during 2023-2025.

Decision-useful implication

The case illustrates why direct-import diversification alone can be an incomplete resilience indicator. A lower recorded share for the restricted country can coexist with greater direct-supplier concentration, continued 100% U.S. net import reliance, and even higher global production concentration. TSDA therefore adds value by separating visible supplier changes from the upstream structure that determines whether dependence actually changed.

Principal sources and reproducibility record

U.S. direct-supplier data: U.S. International Trade Commission DataWeb, HTS 8112.92.1000. Primary production and net import reliance: U.S. Geological Survey Mineral Commodity Summaries. European intermediary data: Eurostat Comext. Canada: Statistics Canada and official/industry processing evidence. Japan: Japan Customs and gallium-specific USGS/JOGMEC evidence. Taiwan: Taiwan Customs Administration. Russia tariff confounder: Public Law 117-110 and the 2022 U.S. Harmonized Tariff Schedule. All calculations, source notes, assumptions, and unresolved classifications are preserved in the TSDA gallium master workbook.

Methodology note: Section 6.4 uses the October 1, 2026 Version 1.1 amendment: "Direct U.S. supplier concentration does not show a robust decline under Section 5." The original empirical-validation rule in Section 8 is retained for this application.